

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.4 Algorithm Execution Observation Table (5 marks)

import math

```
def recursive_fibonacci_count(n):
    if n <= 1:
        return 1
    else:
        return 1 + recursive_fibonacci_count(n-1) + recursive_fibonacci_count(n-2)

def generate_execution_observation_table(sizes):
    result = []
    result.append("Algorithm Execution Observation Table")
    result.append("InputSize RecursiveFactorial IterativeFactorial RecursiveFibonacci  
IterativeFibonacci LinearSearch BinarySearch BubbleSort InsertionSort")

    for n in sizes:
        recursive_factorial = n + 1
        iterative_factorial = n
        recursive_fibonacci = recursive_fibonacci_count(n)
        iterative_fibonacci = n
        linear_search = n
        binary_search = math.floor(math.log2(n)) + 1
        bubble_sort = n * (n-1) // 2
        insertion_sort = n * (n-1) // 2

        result.append(
            f"{n} {recursive_factorial} {iterative_factorial} "
            f"{recursive_fibonacci} {iterative_fibonacci} "
            f"{linear_search} {binary_search} "
            f"{bubble_sort} {insertion_sort}"
        )
    return result
```